

GED Math: Equations, Inequalities, and Functions

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Lesson 1: Solving Linear Equations

- Understand the structure of linear equations.
- Apply inverse operations to isolate variables.
- Check your solution by substitution.

Worked Examples

- ① $2x + 5 = 13 \Rightarrow 2x = 8 \Rightarrow x = 4$
- ② $5x - 3 = 2x + 6 \Rightarrow 3x = 9 \Rightarrow x = 3$
- ③ $7 - 2x = 11 \Rightarrow -2x = 4 \Rightarrow x = -2$
- ④ $3(x + 2) = 15 \Rightarrow x + 2 = 5 \Rightarrow x = 3$
- ⑤ $\frac{x}{4} + 2 = 5 \Rightarrow \frac{x}{4} = 3 \Rightarrow x = 12$

Lesson 2: Solving Inequalities

- Solve inequalities similarly to equations.
- Flip the inequality when multiplying/dividing by a negative number.
- Represent solutions on a number line.

Worked Examples

① $x + 3 < 7 \Rightarrow x < 4$

② $2x - 5 \geq 1 \Rightarrow 2x \geq 6 \Rightarrow x \geq 3$

③ $-3x \leq 9 \Rightarrow x \geq -3$ (flip inequality)

④ $5 - x > 2 \Rightarrow -x > -3 \Rightarrow x < 3$

⑤ $\frac{x}{2} + 4 \leq 6 \Rightarrow \frac{x}{2} \leq 2 \Rightarrow x \leq 4$

Lesson 3: Systems of Linear Equations

- Solve by substitution or elimination.
- Interpret intersection point as common solution.

Worked Examples

① $x + y = 10, y = 4 \Rightarrow x = 6$

② $2x + y = 7, x - y = 1 \Rightarrow (x, y) = (\frac{8}{3}, \frac{5}{3})$

③ $3x - y = 5, 2x + y = 4 \Rightarrow (x, y) = (\frac{9}{5}, -\frac{2}{5})$

④ $y = 2x + 1, y = -x + 4 \Rightarrow (x, y) = (1, 3)$

⑤ $x + 2y = 8, 3x - y = 5 \Rightarrow (x, y) = (3, 2.5)$

Lesson 4: Functions and Their Representations

- A function assigns exactly one output to each input.
- Use function notation: $f(x) = 2x + 3$.
- Identify domain and range.

Worked Examples

- ① $f(x) = 2x + 3, f(4) = 11$
- ② $g(x) = -x^2 + 4, g(2) = 0$
- ③ $h(x) = 3x - 5, h(-1) = -8$
- ④ $f(x) = |x - 2|, f(5) = 3$
- ⑤ Determine if $\{(1, 2), (2, 3), (3, 4), (3, 5)\}$ is a function — No (3 maps to two values)

Lesson 5: Quadratic Functions

- Standard form: $y = ax^2 + bx + c$
- Axis of symmetry: $x = -\frac{b}{2a}$
- Vertex: (h, k)

Worked Examples

- ① $y = x^2 - 4x + 3 \Rightarrow$ vertex at $(2, -1)$
- ② $y = -x^2 + 2x + 3 \Rightarrow$ vertex at $(1, 4)$
- ③ $y = 2(x - 1)^2 + 3$ vertex form $(1, 3)$
- ④ Solve $x^2 - 5x + 6 = 0 \Rightarrow (x - 2)(x - 3) = 0 \Rightarrow x = 2, 3$
- ⑤ Axis of symmetry of $y = x^2 + 6x + 5$ is $x = -3$